

Challenge 7

Pre/Post Increment and Decrement



<pre>#include <stdio.h> int main() { int a = 5; printf("%d\n", a++); printf("%d\n", a); return 0; }</pre>	<pre>#include <stdio.h> int main() { int a = 5; printf("%d\n", ++a); printf("%d\n", a); return 0; }</pre>	<pre>#include <stdio.h> int main() { int a = 10; printf("%d\n", a--); printf("%d\n", a); return 0; }</pre>
<pre>#include <stdio.h> int main() { int a = 5; int b = a++ + ++a; //Undefined behaviour printf("a=%d\n", a); printf("b=%d\n", b); return 0; }</pre>	<pre>#include <stdio.h> int main() { int a = 5; int b = 10; int c = a++ + ++b; printf("a= %d\n",a); printf("b= %d\n",b); printf("c=%d\n", c); return 0; }</pre>	<pre>#include <stdio.h> int main() { int a = 5; int b = ++a + 10; printf("a=%d\n", a); printf("b=%d\n", b); return 0; }</pre>
<pre>#include <stdio.h> int main() { int a = 5; int b = a++ * 2; printf("a = %d\n", a); printf("b = %d\n", b); return 0; }</pre>	<pre>#include <stdio.h> int main() { int a = 5; int b = ++a * 2; printf("a = %d\n", a); printf("b = %d\n", b); return 0; }</pre>	<pre>#include <stdio.h> int main() { int a = 5; a++; ++a; a--; printf("%d", a); return 0; }</pre>

```
#include <stdio.h>
```

```
int main() {  
    int a = 5;  
    int b = a++;  
  
    b++;  
  
    printf("a = %d\n",  
a);  
    printf("b = %d\n",  
b);  
  
    return 0;  
}
```

```
#include <stdio.h>
```

```
int main() {  
    int a = 5;  
  
    printf("%d ", a++);  
    printf("%d ", ++a);  
    printf("%d ", a--);  
    printf("%d", --a);  
  
    return 0;  
}
```

```
#include <stdio.h>
```

```
int main() {  
    int a = 10;  
  
    int b = a++ - --a;  
  
    printf("a = %d\n",  
a);  
    printf("b = %d\n",  
b);  
  
    return 0;  
}
```

GATE Questions



```
#include <stdio.h>
```

```
int main()  
{  
    int m = 10;  
    int n, n1;  
  
    n = ++m;  
    n1 = m++;  
  
    n--;  
    --n1;  
  
    n -= n1;  
  
    printf("%d", n);  
  
    return 0;  
}
```

```
#include <stdio.h>
```

```
int main()  
{  
    int a = 2;  
    int b, c;  
  
    b = 2 * (a++);  
    c = 2 * (++a);  
  
    printf("b = %d\n",  
b);  
    printf("c = %d\n",  
c);  
  
    return 0;  
}
```

```
#include <stdio.h>
```

```
int main()  
{  
    int i = 0, count =  
0;  
  
    for (int j = -3; j  
<= 3; j++)  
    {  
        if ((j >= 0) &&  
(i++))  
            count += j;  
    }  
  
    count += i;  
  
    printf("%d", count);  
  
    return 0;  
}
```

More Questions



```
#include <stdio.h>
```

```
int main()
{
    int i = 0;

    if (1 || i++)
        printf("A");

    printf(" %d", i);

    return 0;
}
```

```
#include <stdio.h>
```

```
int main()
{
    int i = 0;

    if (0 || i++)
        printf("A");

    printf(" %d", i);

    return 0;
}
```

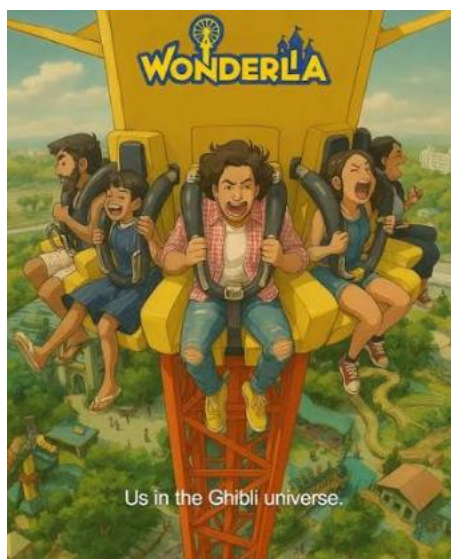
```
#include <stdio.h>
```

```
int main()
{
    int i = 0;

    if (1 && ++i)
        printf("A");

    printf(" %d", i);

    return 0;
}
```



```
#include <stdio.h>

int main()
{
    int i = 0;

    if (0 && ++i)
        printf("A");

    printf(" %d", i);

    return 0;
}
```

```
#include <stdio.h>

int main()
{
    int i = 0;
    int count = 0;

    for (int j = 0; j < 5; j++)
    {
        if (j % 2 == 0 && i++)
            count++;
    }

    printf("%d %d", i, count);

    return 0;
}
```

```
#include <stdio.h>

int main()
{
    int i = 0;
    int count = 0;

    for (int j = 0; j < 5; j++)
    {
        if (j % 2 != 0 || i++)
            count++;
    }

    printf("%d %d", i, count);

    return 0;
}
```

```
#include <stdio.h>

int main()
{
    int i = 0;
    int sum = 0;

    for (int j = 1; j <= 5; j++)
    {
        if (j < 3 && ++i)
            sum += j;
    }
}
```

```
printf("%d %d", i, sum);

return 0;
}
```

```
#include <stdio.h>

int main()
{
    int i = 0;
    int count = 0;

    for (int j = 1; j <= 5; j++)
    {
        if ((j < 3) || (i++))
            count++;
    }

    printf("%d %d", i, count);

    return 0;
}
```

```
#include <stdio.h>

int main()
{
    int a = 5;
    int b = 0;

    if (a++ > 5)
        b++;

    if (++a > 6)
        b++;

    printf("%d %d", a, b);

    return 0;
}
```

```
#include <stdio.h>

int main()
{
    int a = 5;
    int b = 0;

    if (a++ == 5 || ++a == 7)
        b++;
}
```

```
printf("%d %d", a, b);
```

```
return 0;
```

```
}
```



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i = 0;
```

```
    int count = 0;
```

```
    for (int j = 0; j < 5; j++)
```

```
    {
```

```
        if ((j % 2 == 0 && i++) || (++i > 3))
```

```
            count++;
```

```
    }
```

```
    printf("%d %d", i, count);
```

```
    return 0;
```

```
}
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i = 0;
```

```
    int count = 0;
```

```
    for (int j = 1; j <= 6; j++)
```

```
    {
```

```
        if ((j % 2 && i++) || (j % 3 == 0 && ++i))
```



```
        count++;  
    }  
    printf("%d %d", i, count);  
    return 0;  
}
```